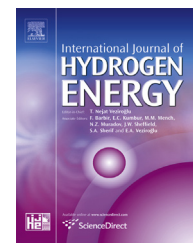


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Pt/Fe-NF electrode with high double-layer capacitance for efficient hydrogen evolution reaction in alkaline media

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ABSTRACT

The electrode with high catalytic activity, low hydrogen overpotential and low cost is desired for hydrogen evolution reaction (HER) via electrocatalytic water splitting. In this study, Pt/Fe-Ni foam (Pt/Fe-NF) electrode was synthesized via cathodic electrodeposition followed by impregnation deposition. Physical and electrochemical properties of Pt/Fe-NF, NF and Pt/NF electrodes were characterized by various techniques. The Pt/Fe-NF electrode exhibited better electrochemical activity for HER under alkaline condition than those of Pt/NF and NF electrodes, owing to the introduction of zero valences Pt and Fe onto the NF, and synergetic effect resulted from the formation of Fe-Ni alloy. Furthermore, Pt/Fe-NF electrode showed extremely high double-layer capacitance (69.1 mFcm^{-2}), suggesting high active sites for the Pt/Fe-NF. Tafel slope of Pt/Fe-NF was 59.9 mV dec^{-1} , indicating that the Volmer-Heyrovsky HER mechanism was the rate-limiting step. The Pt/Fe-NF electrode with great electrocatalytic activity is a promising electro-catalyst for industrial hydrogen production from alkaline electrolyte.

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Introduction

Hydrogen, as a renewable and clean fuel, represents one of the most promising new energy sources [1,2]. Water electrolysis is an important method for the production of highly pure hydrogen [3]. At present, the most commonly used catalyst for HER is platinum (Pt), since it has high catalytic activity and very low over-potential for HER [4,5]. However, as a noble metal, its depletion and high cost hinders its practical application. Hence, efficient alternatives or reducing the usage of Pt are of great importance. It was found that the

earth-abundant metals or their compounds, such as Ni, Co, Fe, Mo, Mo-P and Mo_2C , could be used as electro-catalysts [4,6–8]. Among these materials, Ni and its alloys, for example, Ni-Fe, Ni-Cu, Ni-Mo, Ni-Zn, and Ni-Co-Zn [9–12] have exhibited high chemical stability and excellent catalytic properties towards HER. It has been demonstrated that the effective catalytic activity of some transition-metals (such as Fe, Ni) for HER could be attributed to Brewer-Engel valence bond theory [13]. That is, whenever metals of the left half of the transition series in the periodic table with empty or less-filled d-orbitals are alloyed with metals of the

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right half of the series with more filled d-bands, a maximum in bond strength and stability of the intermetallic alloy phases is expected, and there is a well-pronounced synergism in the electrocatalysis, resulting in good electron-transport properties as well as the efficient absorption and desorption of hydrogen in the process of hydrogen evolution [13]. In other words, the composite catalyst forms the synergetic effect, which is advantageous to HER. Therefore, transition-metals are promising materials for HER.

Despite of the HER catalysts, conductive supporting are also essential in electrochemical systems. The supporting material is a crucial factor in determining the particle size and distribution of the metal particles located on its surface, which has a great influence on the amount of active sites, catalytic performance and stability of the supported catalyst [14,15]. The reported supporting materials include multi-walled carbon nanotubes [16], reduced graphene oxide [17], glassy carbon [18], stainless steels [19], and NF [20] and so on. Among these conductive substrates, NF is an ideal cathode substrate since it possesses good electrical conductivity, high specific surface area, uniform pores, high H₂ uptake capacity and good mechanical strength [21–25].

The electrodes are usually prepared by loading the catalysts onto the supporting materials, for instance, Pt/Ni [26] and Cu-Pt/NPCC [27]. However, preparation of the advanced Pt/Ni electrode not only required a large amount of Pt but also fairly long reaction time [26]. Introduction of Cu into the catalyst could substantially reduce the amount of Pt; whereas, the standard potential of Cu (0.337 V vs. NHE) was relatively high, resulting in high hydrogen evolution overpotential and impeding HER on the electrode surface. Thus, the transition metal with lower standard potential would be more satisfied as doping material. As mentioned above, Fe, which has been used as electro-catalyst material, has a low standard potential of −0.440 V vs. NHE; furthermore, the less-filled 3d-orbitals of Fe (the electronic track arrangement of 3d⁶4s²) would be paired with the 1s hydrogen atom to form Fe-H during HER. Meanwhile, as it is well known that the M–H bond energy (where M represents various metal elements) exhibits a ‘volcano’-shaped curve on heterogeneous catalysis [28]. Hydrogen adsorption energy that is too high would prevent the release of H₂ at the surface of catalyst. In contrast, hydrogen adsorption energy that is too low would result in slow electron-transfer rates. According to the ‘volcano’-shaped curve, the Fe-H bond energy is moderate [28], which is favorable to HER. Hence, it is hypothesized that the combination of Pt, Fe and NF could be a promising cathode material for HER.

Herein, in this report, we attempted to synthesize Pt/Fe-NF electrode using the electrodeposition technique and impregnation deposition method. The Pt/Fe-NF electrode was expected to have the following superiorities: (a) catalytic activity could be enhanced since both efficient Pt with low overpotential and promising transition-metal Fe were loaded onto the 3D porous, dendritic structured NF; (b) the addition of Fe is beneficial to the Pt loading on NF electrode; (c) the original usage amount of Pt could be significantly reduced as composite material was prepared, which decreased the overall cost of the electrode for HER. The performance of the as-prepared electrodes were determined in terms of onset

potential, double-layer capacitance, charge-transfer resistance, Tafel slopes and stability, and so on.

Experimental

Reagents and materials

Chloroplatinic acid (H₂PtCl₆), Na₂SO₄ and ethanol (C₂H₅OH) were purchased from Guangzhou Jinhua Chemical Reagent Co. Ltd. (Guangzhou, China); NaBH₄, H₂SO₄, KOH, and FeSO₄·7H₂O were purchased from Shantou Xilong Chemical Factory (Shantou, China); Hexadecyl trimethyl ammonium bromide (CTAB) were purchased from Aladdin Ltd; NF (pores per linear inch: 120 ppi, apparent density: 0.1–0.8 g cm^{−3}, void volume of the alloy: 60%–98%) were purchased from Suzhou Jialongde Foam Metals Co. Ltd. (Suzhou, China). All chemicals were of analytical grade and directly used without further purification. The Ultra-pure water (18.25 MΩ cm) was used for solution preparation.

Preparation of Pt/Fe-NF electrode

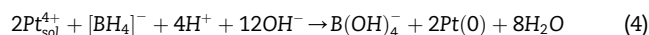
Pt/Fe-NF electrode was prepared by a two-step procedure, including: (a) cathodic electrodeposition preparation of Fe deposited NF (Fe/NF) electrode, (b) impregnation deposition preparation of Pt/Fe-NF electrode.

- (a) Cathodic electrodeposition. Fe/NF electrode was synthesized using the electrodeposition technique [29]. This method could not only remove the use of polymer binders but also result in smoother surface and better nanoparticles dispersion on the electrode [29,30]. Briefly, prior to the electrodeposition, NF electrode (5.0 cm × 3.0 cm × 0.17 cm) was firstly steeped with H₂SO₄ (1 M) for 5 min to the remove surface native oxides; then, NF electrode was washed respectively with C₂H₅OH and Ultra-pure water for 15 min under ultrasonic condition to remove the surface impurities. After that, the pretreated NF electrode was used as the cathode and graphite (5.0 cm × 3.0 cm × 0.4 cm) electrode was served as the anode, and the electrodeposition was carried out for 30 min under a constant current of 10 mA condition, and with a solution of 15 mM FeSO₄·7H₂O, 10 mM Na₂SO₄ and 1 mM CTAB as the electrodeposition electrolyte. The addition of Na₂SO₄ was to improve the conductivity of the solution, while adding CTAB was to reduce the agglomeration of nanoparticles [31]. Then, the Fe/NF electrode was obtained. Fe was deposited on the surface of NF electrode according to Eq (1).



- (b) Impregnation deposition. The as-prepared Fe/NF electrode was submerged in 50 mL of 0.6 mM aqueous H₂PtCl₆ solution in a beaker and stirred with 120 r min^{−1} for 4 h at 30 °C. After soaking for 4 h in the Pt salt solution, the electrode was then soaking in 10 mL of 0.918 M NaBH₄ solution (initial pH was adjusted to 12

using NaOH) for 30 min. Then the as-prepared electrode was washed with Ultra-pure water and dried at 30 °C under vacuum. Pt would deposit on the surface of the Fe/NF electrode by Eqs. (2)–(4).



For comparison, Pt/NF electrode was also used in the experiments. Pt/NF electrode was prepared by impregnation deposition method, the same as the second step of Pt/Fe-NF electrode synthesis.

Physicochemical characterizations

The morphologies of the as-prepared electrodes were characterized by scanning electron microscopy (SEM, Hitachi S-3400N), and the respective compositions were investigated by energy-dispersive X-ray spectroscopy (EDS). The phase structure was analyzed by X-ray diffraction (XRD) using a diffractometer (Rigaku D/MAX 2500 V, Japan) with Cu K α ($\lambda = 1.54056$) radiation flux at a scanning rate of 5° min⁻¹ in the 2 θ range of 20–90°. And the accelerating voltage and applied current were 40 kV and 200 mA, respectively.

Inductively coupled plasma-atomic emission spectroscopy (ICP-AES, CAMT 2000 Thermo Fisher Scientific) was utilized for quantitative determination of Pt loading amount on the electrodes. Liquid samples for ICP-AES were prepared through the following steps: (a) Pt/NF (0.5030 g) and Pt/Fe-NF (0.5850 g) electrodes were weighed using an analytical balance; (b) each electrode was digested in 15.0 mL of aqua regia (mixture of HCl and HNO₃ with 3:1 proportion by volume) and heated at 120 °C; (c) after the samples were completely digested, they were continuously heated to evaporate the aqua regia, then naturally cooled down to room temperature; (d) the samples were diluted with 5 mL of Ultra-pure water for ICP analysis.

X-ray photoelectron spectroscopic (XPS) was carried out by Thermo ESCALAB 250XI with a monochromatic Al K α X-ray source. Pt/Fe-NF sample was firstly dried in the oven under vacuum, and then deposited on silicon wafer supports for testing. And the obtained data were reported with the binding energy calibrated by C1s as reference energy (C1s = 284.6 eV).

Electrochemical measurements

Electrochemical tests, including linear sweep voltammetry (LSV), cyclic voltammetry (CV), electrochemical impedance spectroscopy (EIS) and the Tafel polarization curves were carried out with an electrochemical workstation (PARSTAT 4000, Princeton Applied Research, USA) at room temperature. Particularly, all electrochemical measurements were conducted in a typical three-electrode cell using NF, Pt/NF or Pt/Fe-NF as the working electrode, a platinum gauze (1 cm × 1 cm) electrode as the counter electrode, and Hg|HgO|OH⁻ (1 M KOH electrolyte) electrode as the reference electrode with 0.05 M KOH solutions, which was purged with N₂ for half an hour before the tests. According to previously

reported literature, hydrogen production by alkaline water electrolysis is the simplest and environmental friendly method [32,33], as compared with acidic water electrolysis since electrolyzer and electrode materials would suffer from corrosion under acidic conditions.

LSV measurement was performed with a scan rate of 20 mV s⁻¹ in the range of 0.3 to -0.8 V vs. Hg|HgO|OH⁻, and the obtained data were reported with iR compensation. CV curves were recorded in the potential range from 0.3 V vs. Hg|HgO|OH⁻ to -0.4 V with various scan rates from 10 to 50 mV s⁻¹ for different electrodes (iR compensation). The durability of the Pt/Fe-NF electrode was assessed by cycling the electrode for 1000 times with each cycle started at 0.10 V and ended at -0.4 V with a scan rate of 50 mV s⁻¹ (iR compensation). EIS were carried out in the frequency range of 100 kHz–0.1 Hz. The obtained experimental EIS data were fitted with an equivalent circuit (2R-CPE circuit model using Zsimpwin). The Nyquist plots were recorded from EIS results. Tafel polarization curves were collected in the potential range from 0.2 V vs. Hg|HgO|OH⁻ to -0.8 V for different electrodes. The double-layer capacitance (C_{dl}) was measured according to the method reported by previous work [34].

Results and discussion

Physicochemical properties of the electrodes

Fig. 1a shows the SEM image of bare NF, which exhibits highly porous reticular structure. This structure is beneficial for mass-transport, owing to its large real surface area [21]. In contrast, surfaces of the Fe/NF and Pt/NF are rough and grainy, leading to relatively large exposed surfaces (Fig. 1b and c). The uniform and dense coating of Fe particles with diameters of around 300 nm on the surface of NF is obvious in Fig. 1b. Fig. 1c presents that the impregnation deposition Pt particles were dispersed on the surface of porous nickel electrode rather than inside the pores; these particles either grew independently or accreted together, with diameters of approximately 200–500 nm.

As can be seen from Fig. 1d, the Pt/Fe-NF electrode surface was covered with a more compact, uneven coating. From the inset of Fig. 1d, which shows the magnified morphology of Fe and Pt particles, cluster-like morphology could be observed. Owing to the small particles deposited on the surface of NF, the effective surface area of Pt/Fe-NF should be increased as compared with the NF electrode. The increase of surface area could be favorable to catalytic activity and beneficial to HER. To characterize the chemical composition of Pt/Fe-NF, EDX was recorded (Fig. 1e). As listed in the inset table, the amounts of Fe and Pt on the Pt/Fe-NF electrode were 22.05 wt% and 2.05 wt%, respectively, confirming the successfully loading of Fe and Pt on NF through the present process.

Pt loading on the electrode is an important parameter since it directly affects the performance of the electrode for HER in electrochemical system. Table S1 presented the theoretical Pt loading and experimental Pt loading on both Pt/NF and Pt/Fe-NF electrodes. As can be clearly seen, though theoretical Pt loading amount for Pt/NF and Pt/Fe-NF electrodes were the same (the theoretical Pt loading was based on the assumption

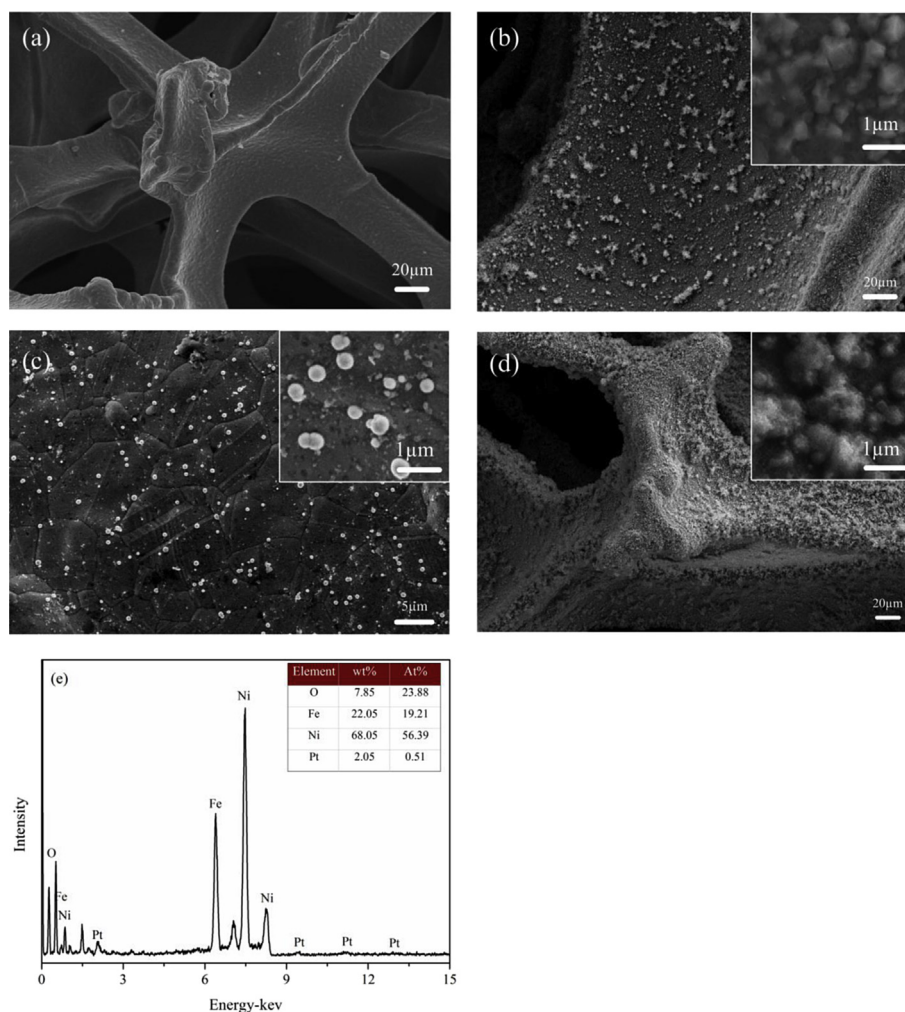


Fig. 1 – SEM images of the electrodes, NF (a), Fe/NF (b), Pt/NF (c), Pt/Fe-NF (d), and (e) Energy dispersive spectrum of Pt/Fe-NF.

that the initial addition of platinum can be fully loaded onto Pt/NF and Pt/Fe-NF electrode), experimental Pt loading on the Pt/Fe-NF ($0.1500 \text{ mg cm}^{-2}$) was higher than that on the Pt/NF ($0.06333 \text{ mg cm}^{-2}$), indicating that introduction of Fe was beneficial for Pt loading on the NF electrode. And this could be part of the reasons that Pt/Fe-NF exhibited better electro-catalytic performance than Pt/NF.

Fig. 2a shows the XRD patterns of NF, Pt/NF and Pt/Fe-NF in the diffraction angle range of $20\text{--}90^\circ$. Three broad peaks appeared at 44.496° , 51.849° and 76.381° were associated with the (111), (200) and (220) planes of Ni, respectively (JCPDS No. 87-0712). Nevertheless, after impregnation deposition, the Pt/NF and Pt/Fe-NF showed no Pt diffraction peaks, and this was likely due to the small loading amount of Pt (as confirmed by EDS and ICP-AES results). Though the content of Fe on the Pt/Fe-NF electrode was relatively high (22.05 wt % according to EDS data), no obvious Fe diffraction peaks were observed. And according to the inset of Fig. 2a, the corresponding peaks shifted slightly to higher angles as compared to those of Pt/NF, indicating a contraction of the lattice upon substitution of Ni with Fe [35] since the radius of the Fe atom (124 pm) is smaller than that of the Ni atom (125 pm). Therefore, a possible explanation for no diffraction peaks of Fe being observed may

be ascribed to the formation of the alloy of Fe and Ni. In comparison with the Pt/NF and Pt/Fe-NF, it is worth noting that the all intensity of Pt/Fe-NF diffraction peak became weaker, which revealed that the addition of Fe possibly changed the crystal form of Ni. In this case, the crystallinity decrease may result from the transformation of the crystal size to the amorphous form. Compared with crystalline materials, amorphous alloys can effectively reduce the activation energy of the adsorption of hydrogen atoms on the surface, thus increase the catalytic activity for HER [36–39].

XPS measurement was employed to study the signals from the elemental compositions of Pt/Fe-NF particles surface. As can be seen from Fig. 2b, the elements of Ni, O, Fe and Pt could be clearly identified on Pt/Fe-NF electrode. Fig. 2c gives the XPS profiles of Ni element in the Pt/Fe-NF. Three metal nickel peaks at about 855.90 eV (Ni $2p_{3/2}$), 858.60 eV (Ni $2p_{3/2}$ sat), and 861.00 eV (Ni $2p_{3/2}$ sat), with six peaks at approximately 855.02 eV (Ni $2p_{3/2}$), 863.00 eV (Ni $2p_{3/2}$ sat), 871.90 eV (Ni $2p_{1/2}$), 873.8 eV (Ni $2p_{1/2}$), 879.10 eV (Ni $2p_{1/2}$ sat) and 881.80 eV (Ni $2p_{1/2}$ sat) could be assigned to the Ni^{2+} and Ni^{3+} species from NiO and Ni_2O_3 . In Fig. 2d, binding energies at 530.30 eV, 531.00 eV and 531.50 eV corresponded to O1s of Fe_2O_3 , NiO and Ni_2O_3 , respectively. Fig. 2e shows the XPS spectrum of the 2p orbit of

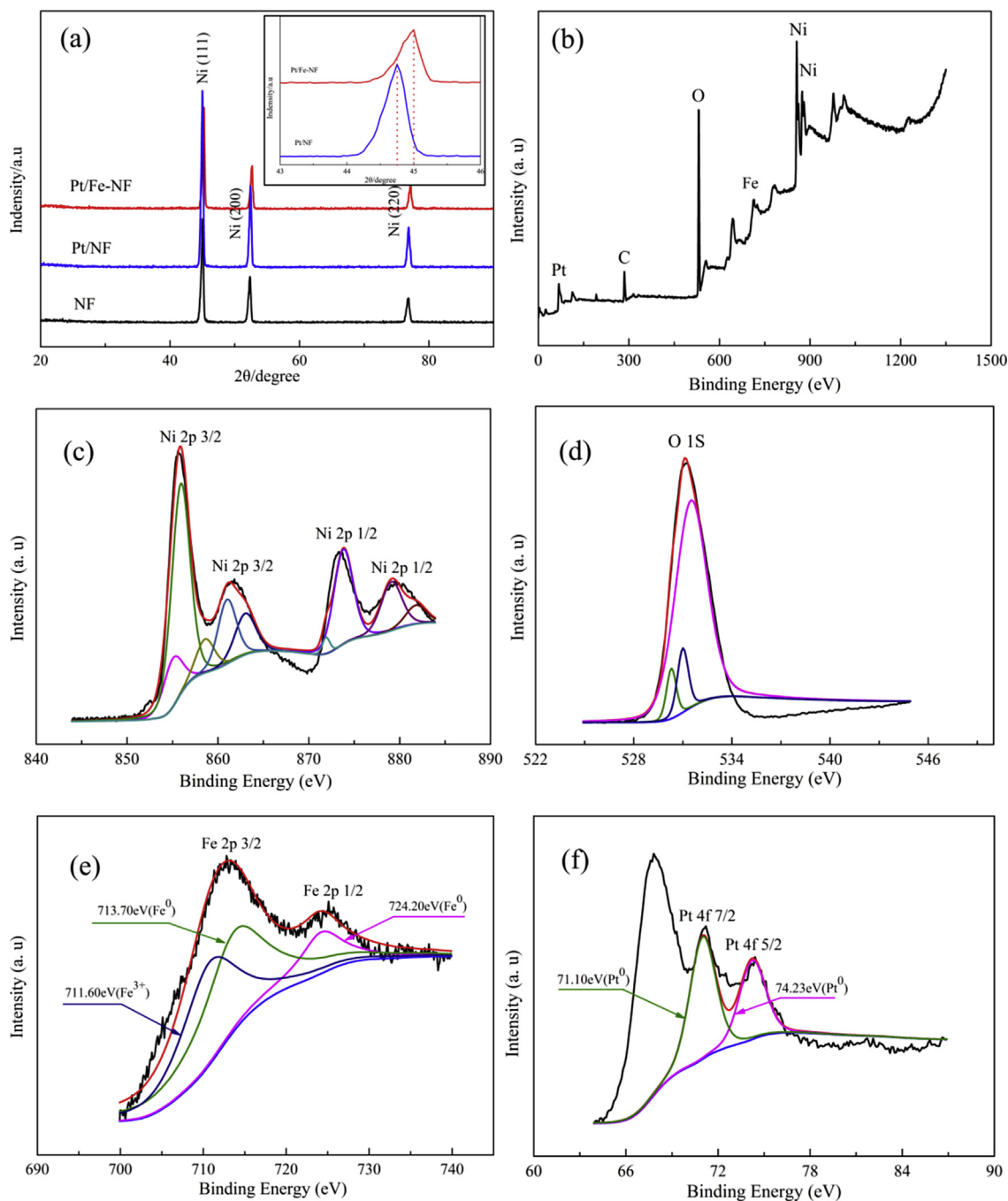


Fig. 2 – (a) XRD patterns of NF, Pt/NF and Pt/Fe-NF electrodes. (b) Pt/Fe-NF XPS survey spectra, (c) High-resolution scans for the Ni, (d) High-resolution scans for the O, (e) High-resolution scans for the Fe, and (f) High-resolution scans for the Pt.

Fe. The 2p core-level spectrum consists of two peaks for metallic Fe at 713.70 (Fe 2p_{3/2}) and 724.20 eV (Fe 2p_{1/2}), and one peak at about 711.60 eV (Fe 2p_{3/2}), which can be assigned to Fe₂O₃. The existence of Fe₂O₃ was likely due to the oxidation of Fe (0) in the air while performing the analysis. Fig. 2f presents the Pt peaks at 71.10 eV (Pt 4f_{7/2}) and 74.23 eV (Pt 4f_{5/2}) for Pt/Fe-NF. Additionally, the Pt 4f peaks show the Pt (0) specie, which is beneficial for HER due to its high catalytic activity and very low over-potential [4,5,40]. The peak at about 68.00 eV, which was shifted beyond Pt binding energy range (>70.00 eV) might

be due to the strong effect of Ni on Pt peak locations. According to previous literature, the content of nickel was an important factor for the Pt 4f peaks shift, and the Pt 4f peaks shifted to a lower binding energy with increasing Ni content in the composite catalyst [35]. Especially, in our case, the NF was used as the supporting material, thus the content of nickel is very high (as revealed by EDS), which could lead to more negative shifts for the Pt 4f peaks. Overall, XPS results demonstrated that the Pt/Fe-NF surface existed Pt (0) and Fe (0) species, which further confirmed that Pt and Fe were

successfully loaded on NF electrode and these zero valence metal can improve the catalytic activity of the catalyst.

Electrochemical properties

To reveal the effects of Pt (0) and Fe (0) species on electrocatalytic activity, electrochemical measurements were carried out. The LSV result in Fig. 3a shows that the Pt/Fe-NF has the smallest onset overpotential ($\eta = -100$ mV) among three electrodes, implying that the catalytic activity of Pt/Fe-NF was the highest. What's more, current densities of the electrodes arranged in an order of Pt/Fe-NF > Pt/NF > NF, indicating that HER can initiate on the surface of Pt/Fe-NF electrodes more easily than Pt/NF and NF electrodes.

The estimation of effective active surface areas of NF, Pt/NF and Pt/Fe-NF electrodes was determined by the C_{dl} in CV curves [41]. Fig. 3b and Fig. S shows CV curves of Pt/Fe-NF, Pt/NF and NF at various scan rates (10, 20, 30, 40 and 50 mV s^{-1}) over the range of 0.3 to -0.4 V vs Hg|HgO|OH $^-$. The C_{dl} of Pt/Fe-NF can be determined by plotting the Δj ($j_a - j_c$) at -0.1 V vs Hg|HgO|OH $^-$ against the scan rate from the CV curve (Fig. 3c), which is expected to be linearly proportional to the effective surface area. The calculated values of C_{dl} were 69.1, 27.6, and

2.30 mF cm^{-2} for Pt/Fe-NF, Pt/NF and NF electrodes, respectively. Obviously, the Pt/Fe-NF exhibited a much higher C_{dl} than the other electrodes, indicating the high exposure of effective active sites. This could be attributed to uniform distribution of the Pt and Fe particles on the electrode surface and the special 3D porous, dendritic structured NF. As a consequence, Pt/Fe-NF possessed rich effective active sites and high overall conductivity, leading to the superior HER activity [42]. The durability of Pt/Fe-NF electrode was also examined. The as-prepared Pt/Fe-NF electrode showed negligible loss even after continuous 1000 cycles (Fig. 3d), indicating the superior long-time stability of the Pt/Fe-NF electrode in alkaline media. All the above results indicated that the Pt/Fe-NF electrode has excellent catalytic activity and stability.

In order to highlight the advantages of the as-developed electrodes for HER and reveal the HER mechanism, Tafel and EIS tests of as-developed electrodes were performed, and the results were shown in Fig. 4. Tafel slope is deemed to be an essential parameter to reveal the inherent property of the catalyst and recognized as the standard for the evaluation of HER [43]. Meanwhile, Tafel plots have been widely used for quantitative kinetics analysis of the HER [44]. Note that for hydrogen evolution in alkaline solutions on a metallic

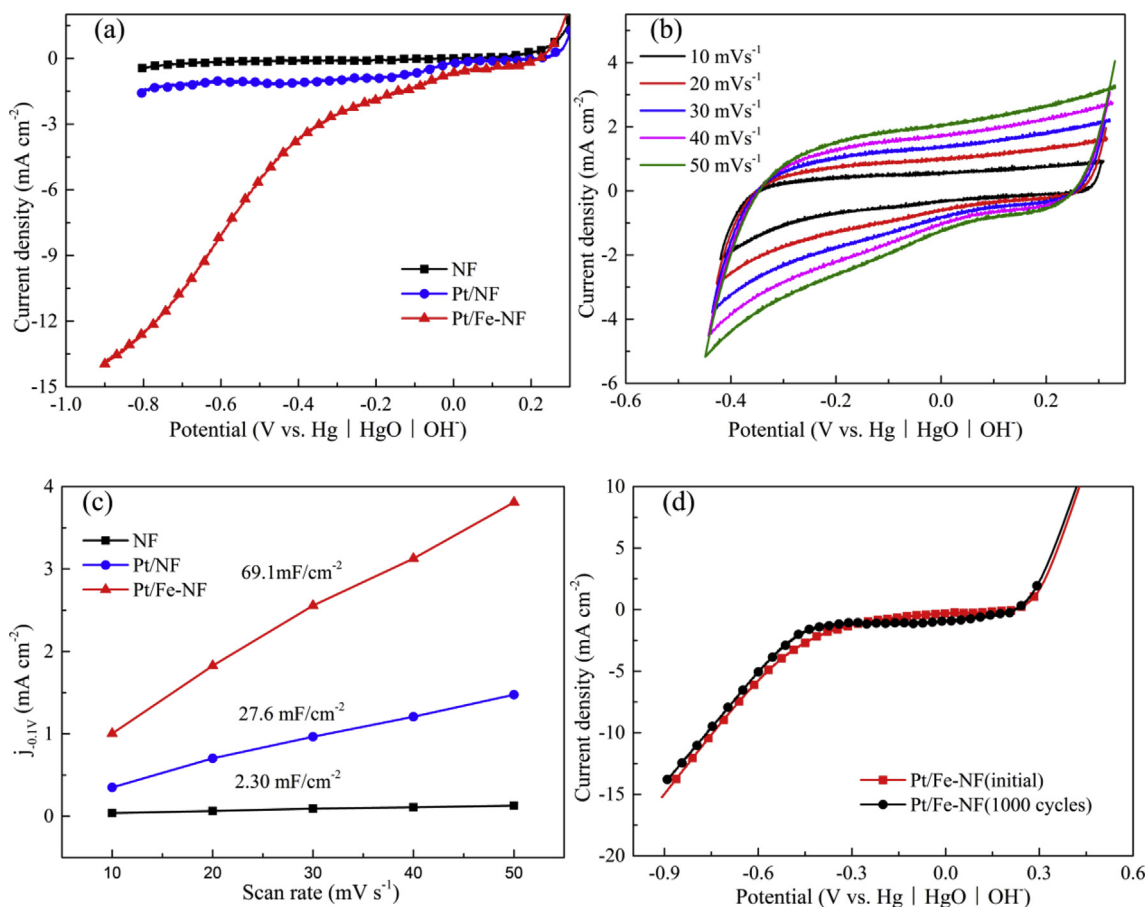


Fig. 3 – (a) LSV (scan rate 20 mV s^{-1}) of NF, Pt/NF and Pt/Fe-NF electrodes in 0.05 M KOH solution (iR compensation); (b) CV for Pt/Fe-NF electrode in 0.05 M KOH solution at different scan rates (iR compensation); (c) Relation of difference between anodic and cathodic currents at -0.1 V with various scan rates from 10 mV s^{-1} to 50 mV s^{-1} (iR compensation), and the slope is double C_{dl} ; (d) Stability test for Pt/Fe-NF electrode in 0.05 M KOH solution at a scan rate of 20 mV s^{-1} (iR compensation).

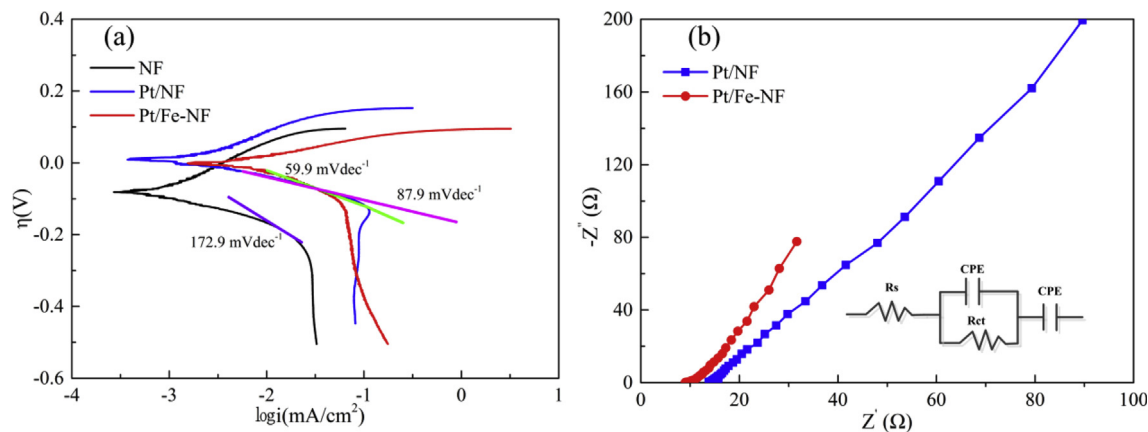


Fig. 4 – (a) Tafel plots of different electrodes in 0.05 M KOH solution; (b) EIS tests of Pt/NF and Pt/Fe-NF electrodes for HER in 0.05 M KOH solution. Inset is the equivalent circuit of system.

electrode M, the mechanism typically involves three major reactions [45]:

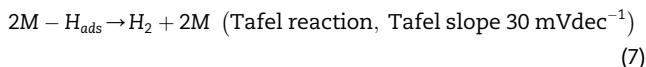
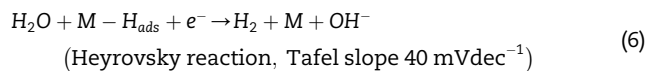
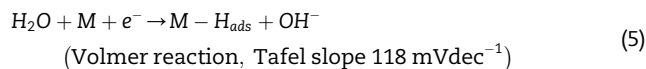


Fig. 4a exhibits the Tafel plots of the NF, Pt/NF and Pt/Fe-NF electrodes, yielding Tafel slopes of 172.9, 87.90 and 59.90 mV dec⁻¹, respectively. Based on the Tafel slopes and involved possible mechanisms, obviously, the value of Tafel slope for Pt/Fe-NF electrode was close to 40 mV dec⁻¹, suggesting the Volmer-Heyrovsky HER mechanism was involved during the HER process [44,46] and the electrochemical desorption step was the rate-limiting step [44,47]. Such small Tafel slope for Pt/Fe-NF electrode could be rationalized as follows: (a) enhanced activity of Pt toward HER due to electronic interactions between Pt and Fe; (b) high catalytic activity resulting from the synergistic effect of Pt, Fe and Ni; (c) improved diffusion of reactants and products on the Pt and Fe surface resulting in more efficient catalytic turnover [48]. Moreover, it has been demonstrated that a smaller Tafel slope means a faster increase of HER rate with the increase of applied potential [49]. The exchange current density (j_0) from Tafel test was used as a parameter to evaluate the activity of the catalysts towards the HER [50]. The calculated j_0 values were $1.494 \times 10^{-3} \text{ mA cm}^{-2}$, $1.692 \times 10^{-3} \text{ mA cm}^{-2}$ and $8.538 \times 10^{-3} \text{ mA cm}^{-2}$ for the NF, Pt/NF and Pt/Fe-NF electrodes, respectively (Table 1). This suggested that the Pt/Fe-NF electrode was more active toward the HER. Therefore, the Pt/Fe-NF electrode could show an outstanding electrocatalytic performance for HER.

EIS is an effective technique to investigate the interface reactions and electrode kinetics of the catalysts in HER [51]. Fig. 4b exhibits the representative Nyquist plots of the EIS of the Pt/NF and Pt/Fe-NF in 0.05 M KOH solution. More detailed analyses were carried out by fitting the impedance data to an

Table 1 – Exchange current densities (j_0) and Tafel slopes of NF, Pt/NF and Pt/Fe-NF electrodes for HER.

Catalyst	Tafel slope (mV dec ⁻¹)	$j_0 \times 10^{-3}$ (mA cm ⁻²)	E_0 (V)
NF	172.9	1.494	-0.07770
Pt/NF	87.90	1.692	0.01667
Pt/Fe-NF	59.90	8.538	0.003282

equivalent circuit (as depicted in the inset of Fig. 4b). The Nyquist plots and fitting results (Table 2) revealed that the charge-transfer resistance (R_{ct}) of Pt/Fe-NF (7.917 Ω) was much lower than that of Pt/NF (30.31 Ω), demonstrating the lower electron transfer resistance or faster electron transfer for Pt/Fe-NF electrode, which was in good agreement with the HER results shown in Fig. 3a. The solution resistances (R_s) for the systems related to Pt/NF and Pt/Fe-NF electrode were 14.69 Ω and 9.893 Ω (Table 2), respectively, manifesting excellent conductivity of electrolyte.

Various analyses and comparisons showed that Pt/Fe-NF electrode outperformed the counterparts of Pt/NF and NF electrodes. This could be attributable to, on the one hand, the addition of Fe was beneficial for Pt loading on the electrode; and on the other hand, and the synergetic effect resulted from the formation of Fe-Ni alloy during the preparation process. In our case, it may easily be assumed that the enhanced performance of Pt/Fe-NF was due to the higher amount of Pt loading on the electrode. However, it was believed that the addition of Fe also played a significant role. As shown in Table S2, though the Pt loading in Cu-Pt/NPCC (28.88 wt%

Table 2 – Fitting results of various resistances of hydrogen evolution system with different electrodes in 0.05 M KOH solution at room temperature.

Catalyst	R_s (Ω)	R_{ct} (Ω)	R_{int} (Ω)
Pt/NF	14.69	30.31	45.00
Pt/Fe-NF	9.893	7.917	17.81

by EDX) was much higher than that of Pt/Fe-NF (2.050 wt% by EDX), the exchange current density (j_0) of the former ($0.2400 \times 10^{-3} \text{ mA cm}^{-2}$) was much lower than the latter ($8.538 \times 10^{-3} \text{ mA cm}^{-2}$). The reason for this could be the formation of Fe and Ni alloy, which leads to the change of electronic structure and exhibits synergistic effect. Meanwhile, there may be another synergistic effect between Fe and Pt [27].

Conclusions

The Pt/Fe-NF electrode was successfully fabricated using electrodeposition and impregnation deposition technique for HER. Various characterization results of the SEM-EDS, ICP-AES, XRD, XPS, LSV, CV, EIS and Tafel curves demonstrated that the Pt/Fe-NF electrode exhibited higher electrocatalytic activity than those of Pt/NF and NF electrodes. This was likely due to, (a) the addition of Fe is beneficial to the Pt loading on NF electrode; (b) the formation of the alloy of Fe and Ni, which played the role of the synergetic effect for HER; (c) its enlarged electrochemical area and rich active sites with high double-layer capacitance. The as-synthesized Pt/Fe-NF electrode with desirable electrochemical activity is a promising electrode material for industrial hydrogen production from alkaline electrolyte.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.ijhydene.2017.02.055>.

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